2] How do PRIMARY KEY and FOREIGN KEY constraints differ?

Ans:

1. Definition

PRIMARY KEY: A constraint used to uniquely identify each record in a table.

FOREIGN KEY: A constraint used to establish a relationship between two tables by linking a column in one table to the PRIMARY KEY in another table.

2. Purpose:

PRIMARY KEY: Ensures that no two rows have the same value in the specified column(s) and that the value cannot be NULL.

FOREIGN KEY: Enforces referential integrity by ensuring that the value in the referencing table matches a value in the referenced table.

3. Number of Constraints:

PRIMARY KEY: A table can have only one PRIMARY KEY, but it can consist of multiple columns (composite key).

FOREIGN KEY: A table can have multiple FOREIGN KEYS referencing different tables.

4. Example:

PRIMARY KEY:

```
CREATE TABLE students (
  student id INT PRIMARY KEY,
  student name VARCHAR (100)
);
```

FOREIGN KEY:

```
CREATE TABLE enrollments (
   enrollment_id INT PRIMARY KEY,
   student_id INT,
   FOREIGN KEY (student_id) REFERENCES students(student_id)
);
```

5. Data Integrity:

PRIMARY KEY: Ensures uniqueness and non-null values within the table.

FOREIGN KEY: Ensures that the referenced value exists in the parent table.